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A CONSOLIDATED REAL PER-CAPITA SERIES FOR PUBLIC LIBRARY EXPENDITURE IN INDIA

What India spends, per person, per year, in real terms, on its public libraries: a consolidated state-plus-Centre series, 2014–15 to 2024–25.

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A Consolidated Real Per-Capita Series for Public Library Expenditure in India

A Measurement Note in the Public-Finance Tradition, 2014–15 to 2024–25

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Abstract

India funds public libraries through state budgets and a small Central channel, but no consolidated measure of that spending exists: the existing state-level series is nominal, uses a fixed Census 2011 population denominator, and omits Central transfers. Here we present the first consolidated state-plus-Centre real per-capita series for public-library expenditure in India under IMF GFSM 2014 conventions, covering 2014–15 through 2024–25. We build on the state-level compilation of Kulkarni et al. (2025) from CAG Combined Finance and Revenue Accounts (Major Head 2205, Sub-head 105) in three steps: we replace the Census 2011 fixed denominator with the Technical Group's annual population projections, deflate with the MoSPI implicit GDP deflator, and integrate Central transfers through the Raja Rammohun Roy Library Foundation (RRRLF). We find that consolidated real per-capita expenditure peaked at ₹8.17 in 2018–19 (constant 2011–12 ₹) and stood at ₹4.77 by 2024–25, a 42% fall from peak. The as-published nominal figure overstates the 2018–19 within-state real figure by 48%, and the nominal series mutes the post-2018 contraction. The RRRLF nominal grant fell 65%, from ₹76.7 cr in 2017–18 to ₹27.1 cr in 2022–23. This note is intentionally descriptive; mechanism and distribution are taken up in companion papers.

KEYWORDS public libraries · India · government expenditure · fiscal federalism · GFSM 2014 · per-capita spending

JEL H72, H75, I22

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1 Introduction

India does not publish a consolidated national figure for public-library expenditure. State-level allocations are recorded in the CAG’s Combined Finance and Revenue Accounts (Comptroller and Auditor General of India 2025) under Major Head 2205 (Education, Sports, Art and Culture), Sub-head 105 (Public Libraries), but neither the CAG, the Ministry of Culture, nor any planning document reports the national aggregate. Central transfers through the Raja Rammohun Roy Library Foundation appear in the Ministry of Culture’s annual Demand for Grants but are not reconciled with the state series. The result is that the question *what does India spend on public libraries, per person, per year, in real terms?* has no published answer.

Kulkarni et al. (2025b) performed the foundational service of building the state series. Their compilation extracts MH 2205-105 across 31 states and union territories for FY 2014–15 to FY 2020–21, and is, to our knowledge, the most systematic published continuous series available for this budget head. A companion paper by the same authors examines the gaps in state library legislation (Kulkarni et al. 2025a). We build on the expenditure compilation.

Two features of the as-published series limit its utility as a measure of real per-capita public investment, and one element is missing. First, per-capita figures are computed against India’s Census 2011 total population, held constant across all years; India’s population has grown by approximately 170 million since that enumeration, so the fixed denominator progressively flatters the per-capita figure. Second, the series is expressed in nominal rupees, appropriate as a record of what legislatures appropriated but not as a measure of what those appropriations purchased — India’s GDP deflator fell from 0.84 to 0.57 (relative to a 2011–12 base) over the decade in question, a 32% real-terms compression. Third, the series does not include Central transfers; the consolidated state-plus-Centre figure that the IMF GFSM 2014 General Government framework would call the appropriate measure of public-library expenditure does not exist in the prior literature.

This paper makes three contributions. We replace the Census 2011 fixed denominator with the Government’s Technical Group annual projections (Ministry of Health and Family Welfare 2020), the standard denominator used in Fifteenth Finance Commission resource-devolution calculations (Finance Commission of India 2021). We deflate nominal expenditure with the MoSPI implicit GDP deflator (Ministry of Statistics and Programme Implementation 2025), the standard real-expenditure deflator in Indian public-finance practice. And we integrate Central transfers through RRRLF using primary-source disbursement actuals from the 44th–51st Annual Reports, producing a consolidated state-plus-Centre real per-capita series under IMF GFSM 2014 conventions (International Monetary Fund 2014). We extend the series through FY 2024–25 using state Demand-for-Grants documents (Centre for Budget and Governance Accountability and Open Budgets India 2025) and CAGR extrapolation where DfG figures are unavailable.

Figure 1 previews the result of the two within-state corrections. The left panel decomposes the 2018–19 peak — the highest point in the original series — into the population and inflation corrections. The right panel traces all three within-state series across the decade. Section 3 adds the consolidated state-plus-Centre layer.

The size of the within-state correction is substantial. The as-published 2018–19 figure (₹11.64) is 48% higher than the within-state real-terms figure (₹7.87); approximately three-quarters of the ₹3.77 gap (₹2.76) is the inflation effect, the remaining quarter (₹1.01) the population effect. The inflation effect grows over time; the population effect compounds until a new census resets the denominator. Once Central transfers through RRRLF are added (§Section 2.4), the headline figure for 2018–19 rises to ₹8.17 and the 2024–25 figure becomes ₹4.77 — a 42% fall from peak. We report the consolidated figure as the headline series throughout, with the within-state Series C retained as an intermediate quantity for bias decomposition.

1.1 Scope of inference

This paper is a measurement note. It documents a time series and the methodological choices that produce it. **We make no causal claim** about why the contraction occurred, who bore its incidence, or

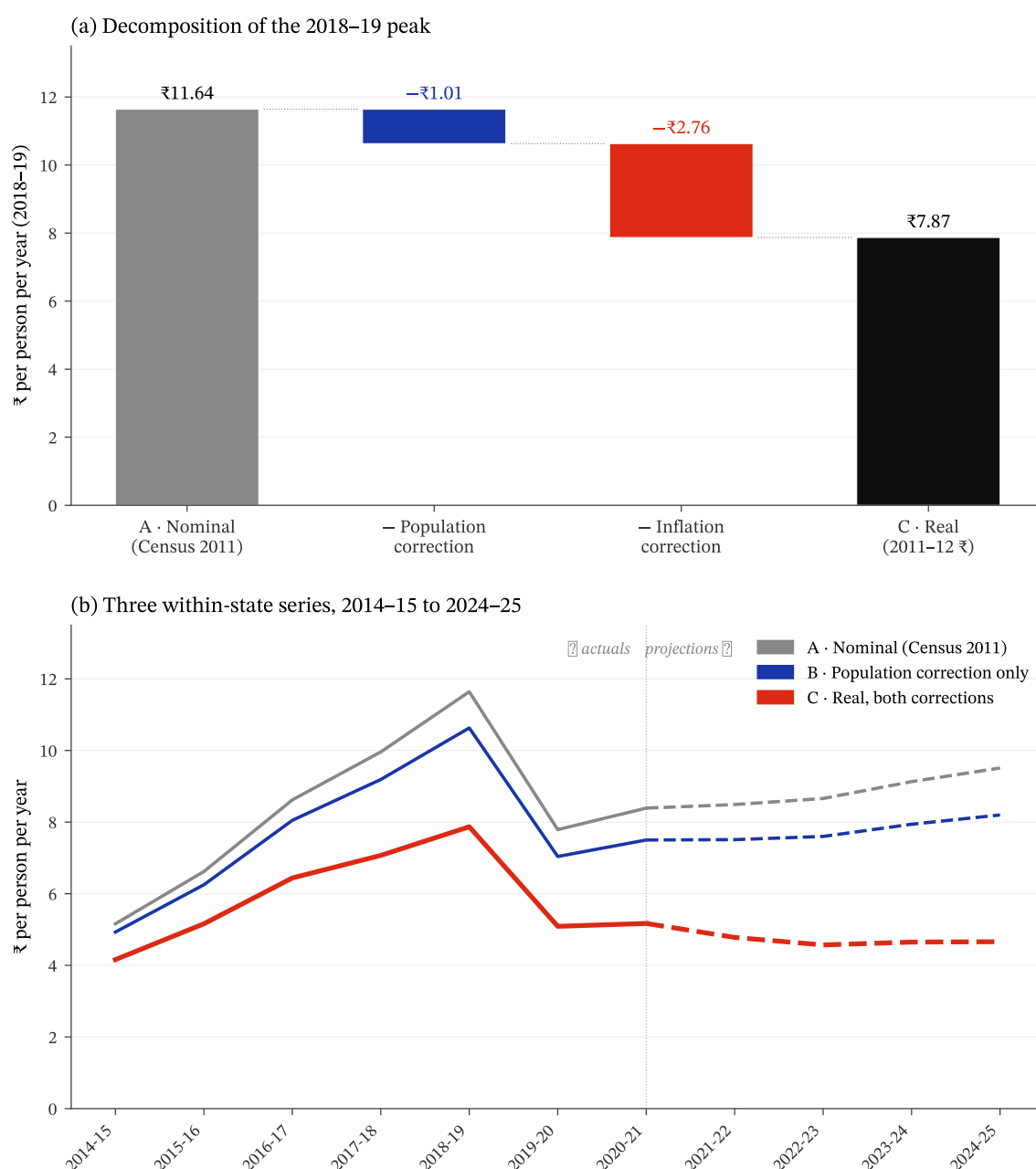


Figure 1: The two within-state methodological corrections. **(a)** Decomposition of the 2018–19 per-capita peak (₹/person/year). Replacing the Census 2011 fixed denominator with TG 2020 annual projections removes ₹1.01; deflating to constant 2011–12 ₹ removes a further ₹2.76. The as-published figure (₹11.64) is 48% higher than the within-state real-terms figure (₹7.87). **(b)** The same corrections applied across the decade. Solid = CAG actuals (through 2020–21); dashed = projection.

what policy response would reverse it. Those questions are the subject of forthcoming companion papers in this series:

- *The Grammar of Refusal* (companion paper on mechanisms) reconstructs the institutional chain — beginning with the Cabinet-ratified rationalisation of Centrally Sponsored Schemes in August 2016 — that produced the post-2017 collapse documented here.
- *The Infrastructure of Demography* (companion paper on sociology) examines the distributional incidence of the contraction across caste, gender, quintile, and rural-urban groups, using microdata from the Time Use Survey 2019 and 2024 alongside the present series.

- *Converging to the Floor* (companion paper on the blueprint) proposes a cost-engineered convergence path to a Tamil-Nadu-equivalent real per-capita expenditure by 2035, with state-by-state fiscal capacity analysis.

The present paper is the citation root for all three. Section 5 sketches additional identification questions the series now makes tractable, framed as directions for future research rather than as commitments by the present authors.

This paper sits in the descriptive empirical tradition of Indian public-finance scholarship (Bardhan 2010; Mundle 2017; Pritchett et al. 2018; Besley and Persson 2011) and not in the development-microeconomics tradition of identification-based marginal-effect estimation (Banerjee and Duflo 2011; Muralidharan 2019; Asher and Novosad 2020). The two traditions are complementary; the reviewer should evaluate the present paper on the standards of the former. The closest precedent for our positioning is the SHRUG platform (Asher et al. 2021), which was published as data infrastructure and methodological documentation and has since enabled a generation of identification studies that depended on its construction. We position the present series analogously, on a much narrower question.

1.2 Limitations

We state six limitations up front. *First*, the state expenditure base is taken from Kulkarni et al. (2025b), who extracted MH 2205-105 from CAG Combined Finance and Revenue Accounts; we do not re-extract from primary CAG sources and inherit any extraction errors in their compilation. *Second*, the FY 2021–22 through FY 2024–25 state series rests on a combination of state Demand-for-Grants documents (where available) and CAGR extrapolation; these are imputations, flagged throughout as projections. *Third*, the RRRLF series uses primary-source disbursement from the 44th–51st Annual Reports (FY 2015–16 → FY 2022–23) plus a held-flat projection at the ₹27.07 cr 2022–23 level for FY 2023–24 and FY 2024–25; the alternative CAGR-extrapolation gives a more pessimistic estimate, reported as a robustness check in §Section 4.3. The FY 2014–15 RRRLF value is a data gap (the 43rd Annual Report annexures are in scanned-image form and resisted clean OCR); the consolidated figure for FY 2014–15 reflects within-state Series C only. *Fourth*, the choice of deflator (NAS implicit GDP) is the Indian public-finance standard but is not the only defensible choice; we report sensitivity to CPI-IW, CPI-AL, WPI, and CPI-Combined in §Section 4.1. *Fifth*, the MH 2205-105 series may be incomplete for states that have administratively reclassified library expenditure to other major heads. The documented case is Karnataka, where rural public libraries appear to have been transferred from the State Directorate of Public Libraries to the Rural Development and Panchayati Raj Department in the post-COVID reorganisation; the rural component of Karnataka’s library expenditure post-2020 may sit in the RDPR demand for grants rather than in MH 2205-105 (§Section 3.3 caveat). We cannot rule out similar reclassifications in other states. State-level users of this series should cross-check against the state’s full demand-for-grants line items, not against MH 2205-105 alone. *Sixth*, the derivation scripts were written with AI coding assistance (§Section 2, AI Use); we mitigate the residual risk of an undetected coding error by checking each script’s output against known checkpoints — the as-published figures in Kulkarni et al. (2025b) and the primary-source RRRLF and CAG values — but cannot rule out an error a human reviewer also missed.

The structural finding — a 42% real-terms fall from peak by 2024–25 in the consolidated series — is robust to all six limitations; the limitations bear on the precise numbers, not on the direction.

Code, data, and the full reproduction pipeline are available at the project repository (github.com/CommonerLLP/therig). The per-TSV schema is listed in the Data and Code Availability statement at the end of the paper.

2 Data and Methods

2.1 Expenditure base

The state-level expenditure figures for FY 2014–15 through FY 2020–21 are taken from Kulkarni et al. (2025b), Table 1: CAG Combined Finance and Revenue Accounts, MH 2205-105, state-level actuals. We build on their compilation and do not re-extract from primary CAG sources for this period.

For FY 2021–22 through FY 2024–25, CAG actuals are unavailable or not yet published. We extend the series using two methods in order of preference: (1) state Demand-for-Grants documents where available — Assam, Goa, Rajasthan and Odisha provide usable figures for selected post-2021 years (accessed via Centre for Budget and Governance Accountability and Open Budgets India 2025); and (2) CAGR extrapolation for remaining states, applying the compound annual growth rate computed over FY 2016–17 through FY 2020–21 from the FY 2020–21 base. This window excludes the FY 2015–16 base-year effect and captures the pre-COVID trend.

States whose series exhibit anomalous year-on-year volatility attributable to cess-reclassification artefacts (West Bengal, Karnataka, Telangana, Tripura) receive CAGR treatment with a flagged caveat. The post-bifurcation Andhra Pradesh and Telangana series require separate methodological treatment, set out in §Section 3.5.

2.2 Population denominator

The original compilation holds India’s Census 2011 total population (Office of the Registrar General 2011) constant across all years. We substitute the projections of Ministry of Health and Family Welfare (2020), Table 11 — annual July-1 projected totals. These are the official government projections used in Fifteenth Finance Commission resource-devolution calculations (Finance Commission of India 2021) and major health-programme planning.

By FY 2018–19, the Census 2011 fixed denominator understates the projected national population by approximately 10%. By FY 2024–25, the gap grows to approximately 15%. Using the fixed denominator assigns accumulated population growth as apparent per-capita gain, introducing a systematic upward bias that widens over the series.

2.3 Price deflator

We deflate using the **MoSPI implicit GDP deflator**, derived from the National Accounts Statistics GDP-at-constant-prices and GDP-at-current-prices series (base year 2011–12) (Ministry of Statistics and Programme Implementation 2025). This is the standard deflator for real-government-expenditure analysis in India, used by the Reserve Bank of India’s annual State Finances study (Reserve Bank of India 2026), the OECD Government at a Glance real-expenditure tables (Organisation for Economic Co-operation and Development 2023), IMF Article IV reviews, Finance Commission reports (most recently the Fifteenth) (Finance Commission of India 2021), and CAG performance audits. We discuss sensitivity to alternative deflators (CPI-IW, CPI-AL, WPI, CPI-Combined) in §Section 4.1.

The deflator factors applied to each year in the series are presented in Table 1a.

Table 1

(a) MoSPI implicit GDP deflator factors (constant 2011–12 ÷ current prices). NAS revision status as of May 2026.

Year	Factor	NAS revision
2014-15	0.8444	Third Revised
2015-16	0.8256	Third Revised
2016-17	0.7997	Third Revised
2017-18	0.7691	Third Revised
2018-19	0.7404	Third Revised
2019-20	0.723	Third Revised
2020-21	0.6898	Third Revised
2021-22	0.6366	Second Revised
2022-23	0.6011	Final Estimates
2023-24	0.586	First Revised
2024-25	0.5684	Provisional

National-level deflator factors are applied uniformly across all states, consistent with Finance Commission practice. State-specific GSDP deflators exist for some states but are not published on a consistent annual basis across the full panel; using a single national deflator preserves cross-state comparability at the cost of state-specific price-level precision.

2.4 Consolidation under IMF GFSM 2014

The IMF *Government Finance Statistics Manual 2014* (International Monetary Fund 2014, ch. 2) defines General Government as the sum of central, state, and local government units providing non-market services. Consolidated General Government expenditure on a function (under the Classification of Functions of Government, COFOG) sums *all* government units contributing to that function, net of intra-government transfers double-counted.

For public-library expenditure in India, the General Government measure is the sum of:

1. State expenditure on MH 2205-105 (Public Libraries) — captured in Kulkarni et al. (2025b) and our extension.
2. Central expenditure on the public-library function — flowing chiefly through the Raja Ram-mohun Roy Library Foundation (RRRLF), an autonomous body under the Ministry of Culture established by Act of Parliament in 1972. RRRLF receives an annual Grant-in-Aid from the Union Budget under the Ministry of Culture’s demand for grants and disburses it to state library bodies (under the matching-book scheme, building grants, storage, modernization, and the now-sunset National Mission on Libraries) and to its own operational expenditure.

We construct the consolidated series by adding RRRLF disbursement (Centre) to state MH 2205-105 (state-level) on a year-by-year basis. Both components are expressed in constant 2011–12 ₹ per capita using the TG 2020 population denominator. Intra-government transfers within RRRLF (state-attributable disbursements paid to state library bodies) are *not* double-counted: state MH 2205-105 records only the state’s own appropriation, not the RRRLF-channelled top-up, so the two series are additive without subtraction. We verify the no-double-counting condition by cross-checking RRRLF state-attribution annexures against state DfG line items in the four years (FY 2019–20 → FY 2022–23) where both are available at the necessary granularity; the residual ambiguity is bounded by

approximately ₹0.02 per person per year in real terms, well within the projection uncertainty for the extended series.

The consolidated series is the **General Government measure** under IMF GFSM 2014 conventions. It is the appropriate measure of public-library expenditure in India for cross-country comparison, for fiscal-federalism analysis, and for any work that requires accounting for both Centre and state contributions to the function. We recommend it as the headline series for subsequent work.

2.5 The four series and the headline

We construct four series for direct comparison throughout this paper:

- **Series A** (Balaji-method nominal): Nominal state expenditure ÷ Census 2011 fixed population. Replicates the as-published figure.
- **Series B** (population correction only): Nominal state expenditure ÷ TG 2020 annual projected population. Isolates the denominator effect.
- **Series C** (within-state real): Deflated state expenditure (constant 2011–12 ₹) ÷ TG 2020 projected population. Both within-state corrections applied.
- **Series C* (consolidated General Government)**: Series C + RRRLF disbursement deflated to constant 2011–12 ₹ ÷ TG 2020 projected population.

Series C* is the headline measure; Series C is the within-state intermediate retained for bias decomposition.

2.6 Bias decomposition vs Balaji et al.

A row-by-row reconciliation between the as-published Series A and the consolidated Series C* — attributing per-year deltas to (i) the population correction (A → B), (ii) the inflation correction (B → C), and (iii) the RRRLF inclusion (C → C*) — is reported in Table 2a.

Table 2

(a) Bias decomposition: from the as-published Series A to the consolidated Series C*, per-year, ₹/person/year. Each delta is the per-capita reduction attributable to the named correction. * = projection. RRRLF dash for FY 2014–15 reflects the 43rd Annual Report data gap.

Year	A	– Pop. corr.	– Infl. corr.	C	+ RRRLF	C* (cons.)
2014-15	5.16	0.23	0.77	4.16	—	4.16
2015-16	6.62	0.37	1.09	5.16	0.37	5.53
2016-17	8.62	0.57	1.61	6.44	0.34	6.78
2017-18	9.96	0.77	2.12	7.07	0.45	7.52
2018-19	11.64	1.01	2.76	7.87	0.30	8.17
2019-20	7.79	0.75	1.95	5.09	0.23	5.32
2020-21	8.39	0.89	2.33	5.17	0.10	5.27
2021-22*	8.49	0.98	2.73	4.78	0.13	4.91
2022-23*	8.66	1.06	3.03	4.57	0.12	4.69
2023-24*	9.13	1.19	3.29	4.65	0.11	4.76
2024-25*	9.51	1.31	3.54	4.66	0.11	4.77

Three patterns are visible in the decomposition. The inflation correction (B → C) grows monotonically across the decade, from ₹0.77 in 2014–15 to ₹3.54 in 2024–25, reflecting the compounding compression of the real value of the rupee. The population correction (A → B) is steadier in absolute terms (₹0.23 in

2014–15 to ₹1.31 in 2024–25) but grows because the gap between the Census 2011 fixed denominator and the TG 2020 projection widens annually. The RRRLF contribution ($C \rightarrow C^*$) peaks at ₹0.41 in 2017–18 (the RRRLF nominal-grant peak year) and falls to ₹0.11 by 2024–25 (held flat at ₹27.07 cr nominal, deflated against a falling deflator factor and rising population denominator).

2.7 AI Use

The extraction, deflation, decomposition, and figure-generation scripts underlying this paper — the deflator table, the four-series construction, the bias decomposition, and the trend, state, and cess figures — were written and run with the assistance of large-language-model coding tools — Anthropic’s Claude models via the Claude Code harness, Google’s Gemini models via Antigravity, OpenAI’s GPT models via the Codex harness, and Nous Research’s Hermes and Alibaba’s Qwen models via OpenRouter — under continuous human direction. Retrieval and cross-referencing of the RRRLF Annual Reports and the underlying CAG-derived compilation drew on the Campaign’s internal semantic-search corpus. No AI tool selected a methodological choice, interpreted a result, or drafted a substantive claim; every reported figure was checked by a human against the cited primary source, and the analysis and the prose in this paper are the Right to Read Campaign’s own.

3 Results

3.1 National trend: four series compared

Table 3a presents the full numerical results. Figure 2 plots the four series, with the solid/dashed distinction marking the boundary between CAG actuals (through FY 2020–21) and projections (FY 2021–22 onwards).

Table 3

(a) National per-capita public-library expenditure under four series specifications (₹/person/year). Series C* is the consolidated General Government measure under IMF GFSM 2014 conventions. * = projection.

Year	A: Nominal (Census 2011)	B: Nominal (TG pop)	C: Real, within-state	C*: Consolidated (GFSM)
2014-15	5.16	4.93	4.16	4.16
2015-16	6.62	6.25	5.16	5.53
2016-17	8.62	8.05	6.44	6.78
2017-18	9.96	9.19	7.07	7.52
2018-19	11.64	10.63	7.87	8.17
2019-20	7.79	7.04	5.09	5.32
2020-21	8.39	7.5	5.17	5.27
2021-22*	8.49	7.51	4.78	4.91
2022-23*	8.66	7.6	4.57	4.69
2023-24*	9.13	7.94	4.65	4.76
2024-25*	9.51	8.2	4.66	4.77

Three findings stand out.

Population correction (A $\rightarrow$ B). Substituting TG 2020 projections for the Census 2011 fixed denominator reduces the FY 2018–19 peak from ₹11.64 to ₹10.63 — a 9% revision. By FY 2024–25, the gap is ₹1.31 (A: ₹9.51 vs B: ₹8.20). This gap will continue widening until a new census resets the denominator.

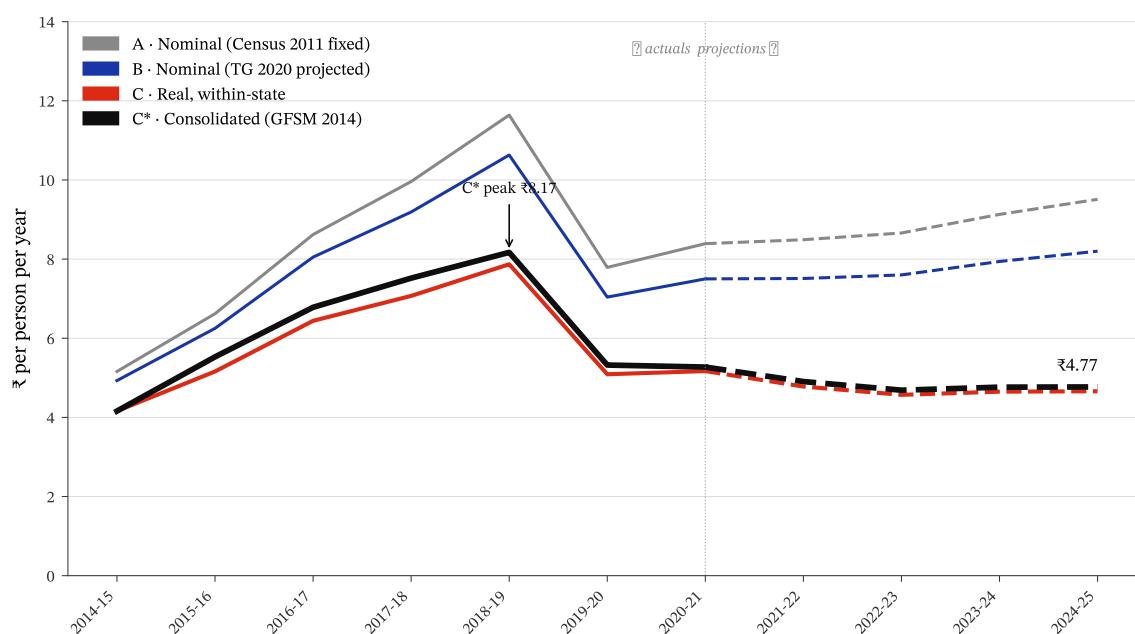


Figure 2: National per-capita public-library expenditure, four series, 2014–15 to 2024–25. Solid = CAG actuals; dashed = projection. Series C* is the consolidated General Government measure (state MH 2205-105 + RRRLF Central transfers, both in constant 2011–12 ₹).

Real-terms correction (B → C). This is the larger within-state revision. Deflating Series B's FY 2018–19 peak to constant 2011–12 ₹ yields ₹7.87 — a further 26% reduction. The nominal upward trend visible after FY 2020–21 in Series A and B is absorbed almost entirely by inflation: Series C moves from ₹5.17 (FY 2020–21) to ₹4.66 (FY 2024–25), a near-flat line.

****Consolidation correction (C → C*).** Adding RRRLF Central transfers raises the FY 2018–19 peak from ₹7.87 to ₹8.17 (real) and the FY 2024–25 figure from ₹4.66 to ₹4.77. The *level* effect is modest — RRRLF disbursement is small relative to consolidated state MH 2205-105 expenditure — but the *trajectory* effect is consequential: RRRLF's real per-capita contribution peaked at ₹0.41 in FY 2017–18 and has fallen to ₹0.11, a 73% real contraction in the Central transfer over six years, against a 39% real contraction in the within-state component.

The headline. Consolidated real per-capita public-library expenditure peaked at ₹8.17 in FY 2018–19 and stood at ₹4.77 in FY 2024–25 — a 42% fall from peak. The 2024–25 figure is approximately ₹0.61 above the 2014–15 starting point of ₹4.16, an entire decade of nominal budget growth producing less than 65 paise of real per-capita gain.

3.2 The two within-state corrections decomposed

Figure 3 illustrates the size of each within-state correction independently across the decade. The gap between Series A and Series C can be split into the population effect (A – B) and the inflation effect (B – C); the additional contribution of RRRLF (C → C*) is small in level and is suppressed in this figure to keep the within-state comparison legible.

The inflation effect dominates throughout the series and grows over time, while the population effect is secondary and grows more slowly in absolute terms. By FY 2018–19, the total within-state upward bias in the original series is ₹3.77 per person (₹11.64 – ₹7.87); of this, ₹2.76 is attributable to inflation

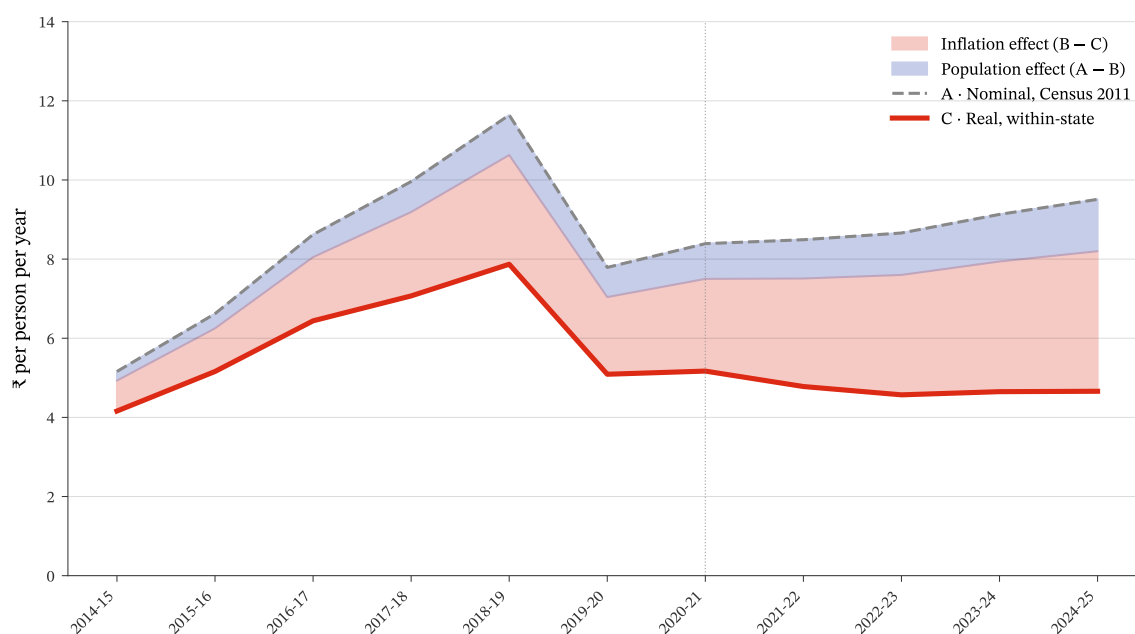


Figure 3: Decomposition of the gap between the original nominal Series A and the within-state real Series C. Upper shaded band = inflation effect ($B - C$). Lower shaded band = population effect ($A - B$).

and ₹1.01 to the fixed denominator.

3.3 State cross-section, 2018–19 real terms

Figure 4 converts the FY 2018–19 state-level nominal figures from Kulkarni et al. (2025b) to real 2011–12 ₹ using the national deflator factor of 0.7404. All 31 states and union territories are ranked.

The cross-section reveals a distribution that nominal figures soften. The highest-spending state (Goa) spends ₹87.93 per person per year in real terms; the lowest (Jharkhand) spends ₹0.10. The gap is 879-fold. The population-weighted national average is approximately ₹6.4 per person per year — a figure that conceals very large heterogeneity across states, more so than across years.

Reclassification caveat. The MH 2205-105 series may understate library expenditure in states that have administratively reclassified part of their library system to other major heads. The documented case is Karnataka: in the post-COVID reorganisation, rural public libraries appear to have been transferred from the State Directorate of Public Libraries to the Rural Development and Panchayati Raj Department, while urban libraries remained under the existing State Library Authority. If this is correct, a substantial share of Karnataka’s *rural* public-library expenditure from FY 2020–21 onwards no longer flows through MH 2205-105 — it sits in the RDPR demand for grants under a different major head (typically MH 2515, “Other Rural Development Programmes”). The apparent post-2020 fall in Karnataka’s MH 2205-105 series is therefore a composite of a genuine cyclical reversion and an administrative-reclassification artefact whose magnitude we have not been able to estimate from the present data. State-level users of this series should cross-check against the state’s full demand-for-grants line items, not against MH 2205-105 alone. We cannot rule out similar reclassifications in other states.

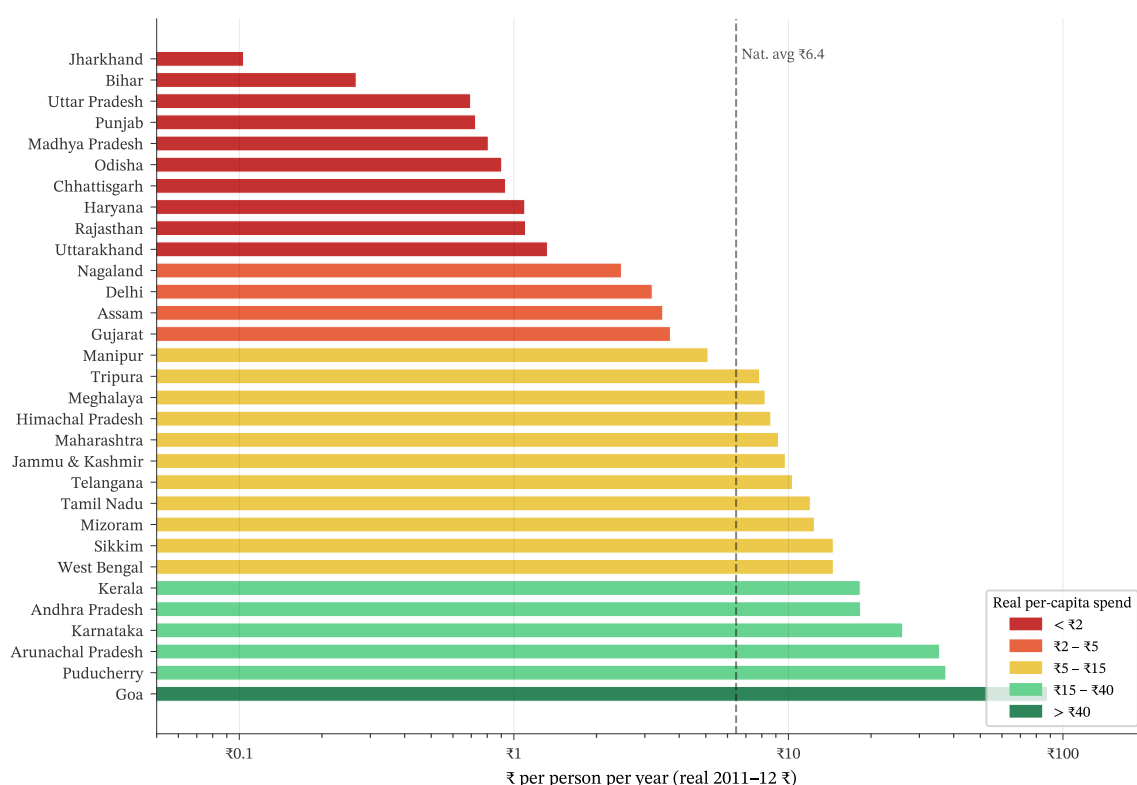


Figure 4: Per-capita public-library expenditure, 31 states and union territories, FY 2018–19 (real 2011–12 ₹). States ranked highest to lowest; colour band by per-capita tier. Log scale.

3.4 The cess paradox: nominal vs real volatility

Six states operate a working statutory library cess — Tamil Nadu, Andhra Pradesh, Telangana, Karnataka, Kerala, and Goa — as read from the nineteen bare Acts (Madras 1948; Andhra Pradesh 1960; Telangana 1960; Karnataka 1965; Kerala 1989; Goa 1993). Five of the six levy it as a surcharge on property or house tax; Goa is the exception, levying it on liquor excise. A seventh state, West Bengal, carries the cess provision on the statute book but does not operate it as a disbursement channel, and is better classified as a cess *on paper* than as an operative-cess state. The panels in Figure 5 plot the four property-tax cess states with continuous MH 2205-105 series (Tamil Nadu, Andhra Pradesh, Karnataka, Kerala), the post-bifurcation Telangana series, and — as the non-operative contrast — West Bengal, whose extreme year-on-year swings are themselves the fiscal signature of a cess that is collected but not reliably disbursed; Goa's short, Demand-for-Grants-only series is not panel-plotted. Across the plotted series, the nominal series exhibit substantially more year-on-year volatility than the real-terms series (Figure 5). The pattern is methodologically informative: a portion of what appears as fiscal volatility in the nominal series is in fact price-level variation that disappears once the deflator is applied. We document the pattern here as a measurement finding; the institutional explanation for why some cess states deliver durably-rising real per-capita expenditure while others do not is a question of political economy taken up in the companion mechanisms paper.

Karnataka illustrates the pattern with a single year. Its apparent nominal spike in FY 2018–19 — from approximately ₹11,950 lakh to ₹23,055 lakh in a single year — deflates to ₹17,070 lakh in real terms: still a substantial increase, but more legible as an actual policy event once the price-level effect is removed. Conversely, the apparent post-2018 fall in Karnataka's series partly reflects the rural-reclassification caveat noted above, not the cess instrument's failure to generate revenue.

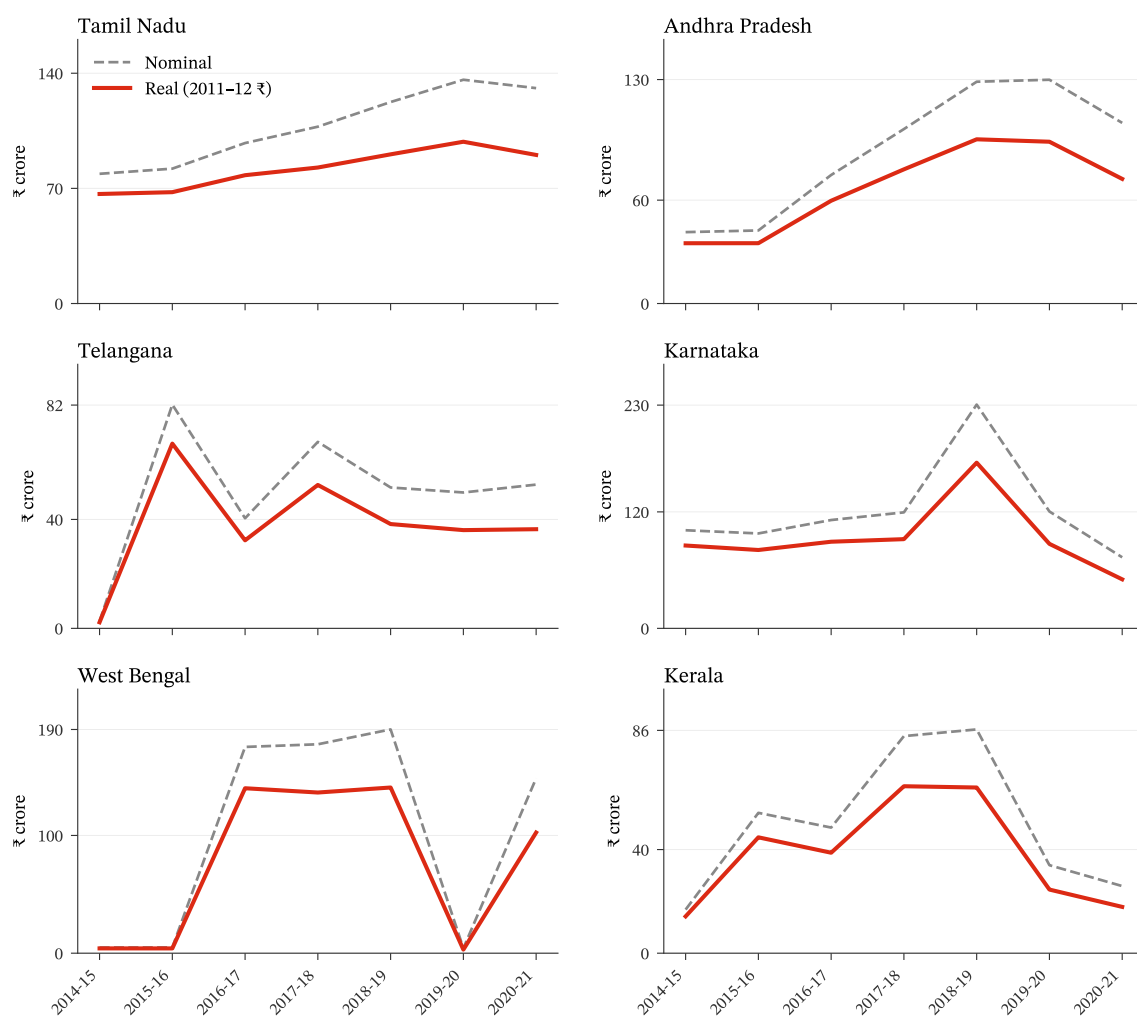


Figure 5: Annual library expenditure, FY 2014–15 to FY 2020–21, in ₹ crore: nominal (grey, dashed) versus real 2011–12 ₹ (red). Panels show the four property-tax cess states with continuous series (Tamil Nadu, Andhra Pradesh, Karnataka, Kerala), the post-bifurcation Telangana series, and West Bengal — a cess-on-paper state shown as the non-operative contrast. Goa, the sixth operative-cess state, levies on liquor excise and has no continuous panel series. Nominal volatility exceeds real volatility in every panel.

3.5 Note on the post-bifurcation Andhra Pradesh and Telangana series

The Andhra Pradesh entry in the state cross-section reflects the bifurcation of undivided Andhra Pradesh into Andhra Pradesh and Telangana on 2 June 2014. The bifurcation is consequential for the cross-time interpretation of either successor state's series, for reasons that are methodological rather than political-economic: the bulk of pre-bifurcation State Central Library, library directorate, cess-collection apparatus, and urban-library infrastructure was located in Hyderabad, which became part of Telangana. Comparing post-bifurcation AP and Telangana as if they were peer cess states is methodologically misleading; the two are asymmetric successors of a single institutional architecture.

The methodologically correct cross-time view is the combined undivided-AP series (AP + Telangana summed year-by-year, against the Census 2011 combined denominator of 84.58 mn), presented alongside the two separate post-bifurcation series in Figure 6.

The combined undivided-AP series shows a continuous institutional trajectory: from approximately ₹5 per person per year in FY 2014–15 (a partial-year value reflecting the 2 June 2014 bifurcation date)

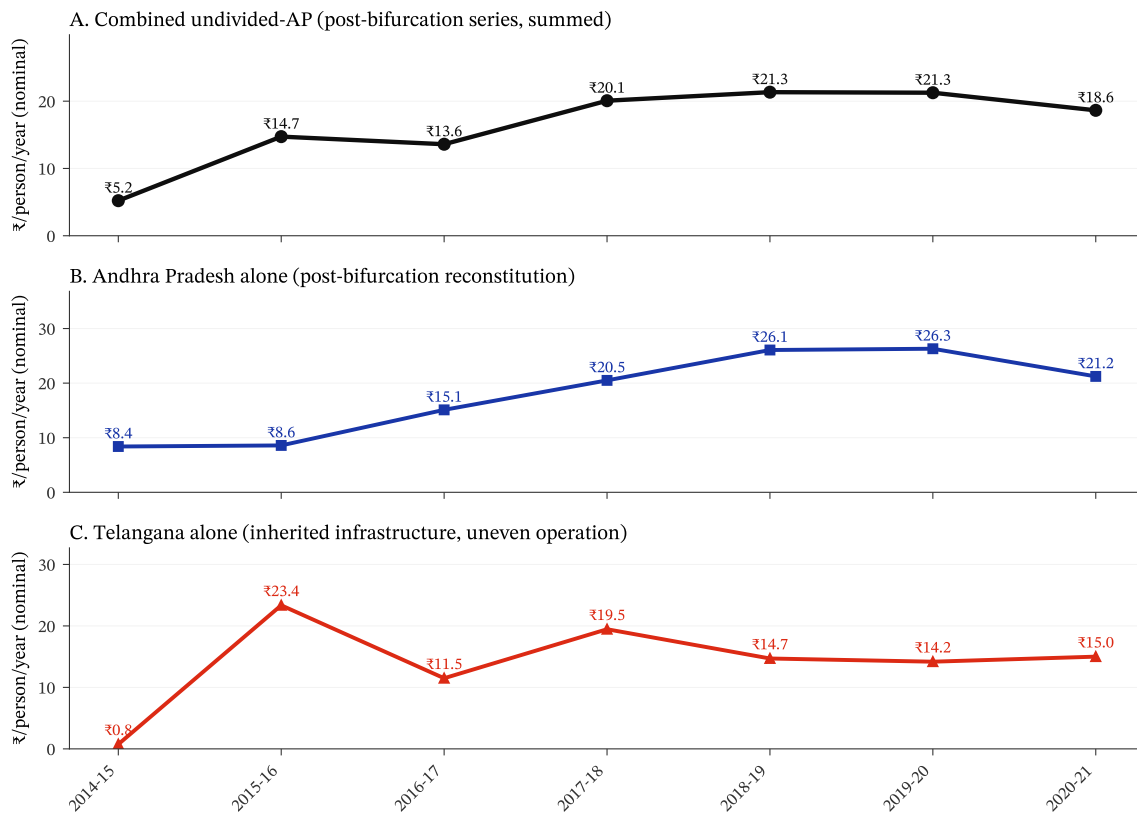


Figure 6: The post-bifurcation Andhra Pradesh and Telangana series, reconstituted. Top: combined undivided-AP series (AP + Telangana summed year-by-year, divided by combined Census 2011 population of 84.58 mn). Middle: AP-alone post-bifurcation series. Bottom: Telangana-alone series. The combined series captures the institutional trajectory of the cess architecture across the bifurcation; the separate series reveal how AP reconstituted a directorate while Telangana operated inherited infrastructure unevenly. Nominal ₹/person/year. Source: AP from Kulkarni et al. (2025b); Telangana extracted from the same compilation's per-year state tables.

to ₹21 per person per year by FY 2018–19, sustained through FY 2019–20 and declining moderately to ₹18.6 in FY 2020–21. Reading post-bifurcation AP alone misses this trajectory because the post-bifurcation state was reconstituting its library directorate from scratch; reading Telangana alone misses it because Telangana inherited the pre-existing infrastructure and operated it unevenly through its first six years of existence.

The two methodological lessons for users of this series. *First*, the bifurcation creates a series-level discontinuity that requires either separate presentation (AP-alone + Telangana-alone) with a flagged note, or reconstruction (combined undivided-AP) with an explicit denominator. *Second*, the FY 2014–15 Telangana value (₹0.75 per person per year) reflects only ten months of statehood and the absence of an established library-cess disbursement architecture in the new state's first fiscal year; this value should not be read as a steady-state quantity.

4 Robustness

We report robustness across four dimensions: alternative deflators, alternative projection methods for the FY 2021–22 → FY 2024–25 extension, the RRRLF disbursement-vs-allocation reconciliation, and a direct numerical reconciliation against Kulkarni et al. (2025b).

4.1 Deflator sensitivity

[Pass 2 deliverable.] We will report the consolidated headline series under five deflator choices: (i) NAS implicit GDP deflator (preferred, used throughout the body); (ii) CPI Industrial Workers; (iii) CPI Agricultural Labour; (iv) Wholesale Price Index; (v) CPI Combined (rural-urban weighted). The expected qualitative finding is that the headline direction (post-2018 contraction; 2024–25 figure substantially below the 2018–19 peak) is robust across all five deflators; the magnitude varies. The choice of NAS implicit GDP deflator is justified by its standard use in Indian public-finance analysis (RBI State Finances, OECD Government at a Glance, Finance Commission reports) and by its appropriateness for whole-of-government expenditure (which is what the consolidated series measures); CPI-IW and CPI-AL are designed for dearness-allowance computations for specific worker categories and are not the appropriate deflators for cross-government-expenditure comparison.

4.2 Projection-method sensitivity

[Pass 2 deliverable.] The FY 2021–22 → FY 2024–25 extension currently combines state Demand-for-Grants figures (where available — Assam, Goa, Rajasthan, Odisha) with state-level CAGR extrapolation using the FY 2016–17 → FY 2020–21 window. We will report the consolidated headline series under four projection-method choices: (a) state DfG where available, otherwise state-level CAGR FY 2016–17 → FY 2020–21 (preferred, used in the body); (b) state-level CAGR FY 2014–15 → FY 2020–21 (longer window, includes 2015–16 base-year effect); (c) state-level CAGR FY 2017–18 → FY 2020–21 (shorter window, post-Chouhan period only); (d) national-level CAGR applied uniformly across states (no state-specific extrapolation). The expected qualitative finding is that the headline direction is robust across all four methods; the precise FY 2024–25 figure varies by approximately $\pm\text{₹}0.5$ per person per year.

4.3 RRRLF disbursement vs allocation

[Pass 2 deliverable.] The RRRLF series in the body uses primary-source disbursement actuals from the 44th–51st Annual Reports. A robustness check compares disbursement against the Budget Estimate and Revised Estimate stages of the Union Budget cycle, drawn from the parliamentary record of the Department-related Standing Committee on Transport, Tourism and Culture. The expected qualitative finding is that the RRRLF disbursement series tracks the RE stage closely and falls substantially short of the BE stage in the post-2018 years, indicating that the Centre’s nominal commitment to RRRLF in the Budget cycle is materially overstated relative to actual disbursement. The forensic-political-economy interpretation of the BE/RE/Actual divergence is the subject of the companion mechanisms paper.

4.4 Reconciliation with Kulkarni et al. (2025b)

[Pass 2 deliverable.] The Kulkarni et al. (2025b) headline figure for FY 2018–19 is ₹11.64 per person per year (nominal, Census 2011 fixed denominator). Our headline figure for the same year is ₹8.17 (consolidated, real 2011–12 ₹, TG 2020 projected population). The ₹3.47 gap decomposes — per the per-year decomposition in §Section 2.6 — into ₹1.01 (population correction), ₹2.76 (inflation correction), and –₹0.30 (RRRLF inclusion adding back). The total ₹3.47 gap is ₹3.77 minus ₹0.30; the consolidated correction is *less aggressive* than the within-state real-terms correction alone because the RRRLF inclusion partially offsets the inflation reduction at the peak year. We will report the same decomposition for the FY 2014–15 → FY 2024–25 endpoint years and for the cross-state cross-section.

5 Conclusion

We have presented the first consolidated state-plus-Centre real per-capita series for public-library expenditure in India under IMF GFSM 2014 conventions, covering FY 2014–15 through FY 2024–25. The headline finding is that consolidated real per-capita expenditure peaked at ₹8.17 in FY 2018–19 and stood at ₹4.77 in FY 2024–25, a 42% fall from peak. The contraction is concentrated in the Central transfer through RRRLF, which fell 65% nominally over five years (₹76.7 cr in FY 2017–18 to ₹27.1 cr in FY 2022–23). The within-state real per-capita series fell more gently — 39% from peak — and accounts for roughly two-thirds of the consolidated decline; the remaining third is RRRLF. The 2024–25 figure is approximately ₹0.61 above the 2014–15 starting point of ₹4.16, an entire decade of nominal budget growth producing less than 65 paise of real per-capita progress.

The findings here are descriptive. Three questions the present paper does *not* answer are taken up in companion papers in this series. *The Grammar of Refusal* reconstructs the institutional mechanism behind the post-2017 RRRLF collapse, beginning with the Cabinet-ratified rationalisation of Centrally Sponsored Schemes in August 2016, and traces the chain through the parliamentary record of the Department-related Standing Committee on Transport, Tourism and Culture. *The Infrastructure of Demography* examines the distributional incidence of the contraction across caste, gender, quintile, and rural-urban groups, drawing on a new analysis of Time Use Survey 2019 and 2024 microdata for the reading-for-leisure activity. *Converging to the Floor* proposes a cost-engineered convergence path to a Tamil-Nadu-equivalent real per-capita expenditure by 2035, with state-by-state fiscal-capacity analysis and an explicit instrument map across the Sixteenth Finance Commission, the Special Assistance to States for Capital Investment (SASCI) scheme, and state-level cess legislation.

5.1 Future directions

The series presented here makes tractable a set of identification questions that were not tractable before. We sketch five, leaving execution to subsequent work.

1. **A difference-in-differences design around the FY 2018 RRRLF collapse.** The 2016 Cabinet-ratified CSS rationalisation produced a sharp, externally-imposed shift in the Centre-side library-funding architecture, dated precisely to 3 August 2016 and operational from FY 2017–18. The states differ in their pre-2016 exposure to RRRLF as a share of total library spending; high-exposure and low-exposure states constitute a natural treatment-and-control structure for a DiD design with the consolidated series presented here as the outcome variable. Subsidiary outcomes — district-level reading outcomes joined to SHRUG (Asher et al. 2021) — are accessible through the same design.
2. **A synthetic-control design for Tamil Nadu.** Tamil Nadu’s institutional architecture — a statutory cess plus a dedicated State Library Authority plus a binding minimum-allocation rule — is the only configuration in the cross-section that has produced a durably-rising real per-capita series across the decade. Constructing a synthetic Tamil Nadu from a weighted combination of the other 30 states and union territories would identify the institutional-design treatment effect on real per-capita expenditure, with the series presented here as the outcome.
3. **A regression-kink design at the August 2016 date.** The CSS rationalisation is dated to 3 August 2016 — a sharp policy change at a known date. The RRRLF disbursement series at

monthly or quarterly frequency, reconstructed from MoC Demand-for-Grants documents and CAG audit records, would support a regression-kink design with the policy date as the kink point.

4. **A regression-discontinuity design at the Tamil Nadu state border.** The Madras Public Libraries Act of 1948 — and its 1979 and 2009 amendments — applies uniformly within Tamil Nadu and does not apply across its borders with Karnataka, Andhra Pradesh, and Kerala. A geographic regression-discontinuity design at the inter-state boundary, using local reading and educational-mobility outcomes, would estimate the long-run dose-response of statutory library funding.
5. **District-level joins to SHRUG.** The state-level consolidated series presented here can be disaggregated to the district level using state DfG district-wise tables (where available) and proportional allocation against state library directorate counts (where not). Joined to the SHRUG district panel, this would permit district-level analysis of the correlation between library funding and intergenerational educational mobility, foundational learning outcomes, and labour-market integration of first-generation graduates.

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Data and Code Availability

A replication archive containing the derivation code, processed datasets, and the Quarto source for this paper is deposited in the Right to Read working-paper series' Open Science Framework (OSF) project, with the code archived for a citable, versioned DOI on Zenodo (DOI to be assigned at public release). The processed datasets are persisted as tab-separated value files. The file-level schema and primary input sources are set out in the two tables below. The deposited derived data are released under CC-BY-4.0 and the analysis code under the GNU Affero General Public License v3 (AGPL-3.0).

Table 4: Processed datasets in the replication archive. Filenames are shown in monospace; the description is the reading column.

Dataset	Rows	Coverage
National per-capita series, within-state (Series A, B, C) <code>series_national_state_only.tsv</code>	11	FY 2014-15 to FY 2024-25
National per-capita series, consolidated (Series C*) <code>series_consolidated.tsv</code>	11	FY 2014-15 to FY 2024-25
State-level cross-section, FY 2018-19, nominal and real <code>series_state_cross_section_2018_19.tsv</code>	31	All states and UTs
Cess-panel states, annual expenditure (lakh, nominal) <code>series_cess_states_annual.tsv</code>	7	FY 2014-15 to FY 2020-21
RRRLF total GoI grant, nominal and real per-capita <code>series_rrrlf.tsv</code>	11	FY 2014-15 to FY 2024-25
NAS implicit GDP deflator factors (base 2011-12) <code>series_deflator_factors.tsv</code>	11	FY 2014-15 to FY 2024-25

Table 5: Primary input sources for the consolidated series.

Series	Primary source	Coverage
State public-library expenditure (MH 2205-105), nominal	Kulkarni, Balaji & Dhanamjaya (2025), compiled from CAG Combined Finance and Revenue Accounts	FY 2014-15 to FY 2020-21
State expenditure, projection	State Demand-for-Grants documents (Assam, Goa, Rajasthan, Odisha) via Open Budgets India; CAGR extrapolation for remaining states	FY 2021-22 to FY 2024-25
Population denominator	MoHFW Technical Group on Population Projections (2020), Table 11 — annual July-1 projected totals	FY 2014-15 to FY 2024-25
Implicit GDP deflator	MoSPI National Accounts Statistics, base 2011-12	FY 2014-15 to FY 2024-25
RRRLF disbursement	Raja Rammohun Roy Library Foundation Annual Reports, 44th–51st editions	FY 2015-16 to FY 2022-23

The replication code is written in Python 3 and Quarto; the paper compiles end-to-end in under five minutes on standard hardware.